STATISTICS · UNIT 4: STATISTICAL INFERENCE

The Later Start Investigation

Use confidence intervals and significance tests to advise the school board on a later start time.

Honors

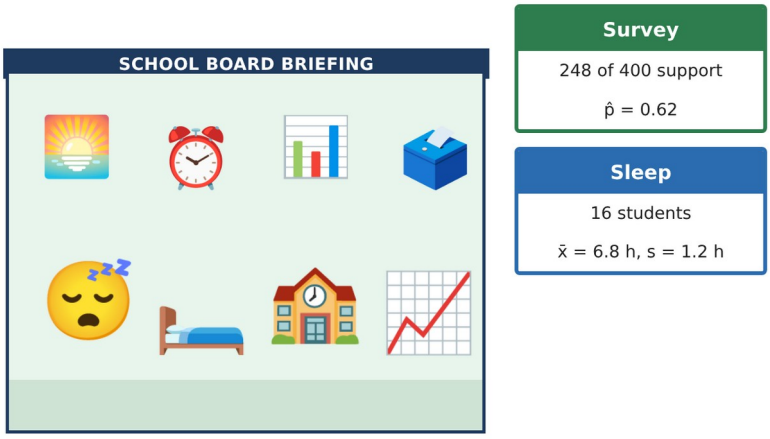
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

The school board is considering starting school 45 minutes later. Board members want to know whether most students support it and whether students are really short on sleep.

Your team will collect evidence with confidence intervals and significance tests and present a recommendation with the right amount of certainty.



Learning Goals

- I can choose, carry out, and interpret one- and two-sample inference procedures.
- I can analyze matched pairs data and explain the role of conditions.
- I can reason about errors, power, and the strength of a recommendation.

Project Timeline

When	What you will do
Day 1	Support for the Change
Day 2	Sleep and the Pilot Week
Day 3	Errors, Power, and Advice
Final	Report to the School Board + reflection

Part 1: Support for the Change

Day 1 · Use the Support survey and By grade cards.

Evidence Cards

Support survey	By grade	Sleep sample
SRS of 400 students	Grade 9: 130 of 200	SRS of 16 students
248 support the change	Grade 12: 118 of 200	$\bar{x}$ = 6.8 h, s = 1.2 h

Doctors say	Critical values	Pilot week
Teens need about	z*: 1.645 (90%), 1.96 (95%)	10 students: extra sleep (h)
8 hours a night	2.576 (99%)	0.5, 1, 0.25, 0.75, −0.25, 1.25, 0.5, 0, 1, 0.5

1. Construct and interpret a 99% confidence interval for the proportion who support the later start.

2. Compare the 95% and 99% intervals. Why is one wider?

3. Carry out a two-proportion z-test of H_0: $p_9 = p_{12}$ vs H_a: $p_9 \neq p_{12}$ (pooled $\hat{p} = 0.62$).

4. Find the sample size per grade needed to estimate $p_9 - p_{12}$ within ±0.05 at 95% (use 0.5 for each p).

Part 2: Sleep and the Pilot Week

Day 2 · Use the Sleep sample and Pilot week cards.

5. Test H_0: $\mu = 8$ vs H_a: $\mu < 8$ for the sleep sample and state the conclusion.

6. Pilot week: find the mean and SD of the 10 differences (later start − normal).

7. Carry out a paired t-test of H_0: $\mu_d = 0$ vs H_a: $\mu_d > 0$.

8. Construct a 95% interval for μ_d ($t^* \approx 2.262$) and interpret it in minutes.

Part 3: Errors, Power, and Advice

Day 3 · Write as a statistician advising the board.

9. For the paired test, describe a Type II error and one consequence.

10. Give two ways to increase the power of the pilot study.

11. Explain why the pilot volunteers limit the conclusion and why the paired design was a good choice.

12. Write a three-sentence recommendation to the board that states your evidence and its uncertainty.

Final Product: Report to the School Board

Your final product must include:

- 99% interval and two-proportion test
- Sample-size plan
- One-sample and paired t procedures with conditions
- Error and power analysis
- Report to the school board with a justified recommendation

Reflection

R1. Why is a paired t-test, not a two-sample t-test, correct for the pilot week?

R2. How do a confidence interval and a significance test give different kinds of information to the board?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Intervals and tests are chosen, carried out, and interpreted correctly and support the recommendation. Goes beyond the minimum with a creative, realistic choice.	Intervals and tests are chosen, carried out, and interpreted correctly and support the recommendation.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Report states the evidence, uncertainty, and limits of the study clearly for a non-statistician. Polished, organized, and visually engaging.	Report states the evidence, uncertainty, and limits of the study clearly for a non-statistician. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.